

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 6: Security, Privacy and Data Integrity

Subtopic 6.1: Data Security

Concept 6.1.2: System Security Measures (Authentication, Firewall, Anti-Virus)

Total Questions: 12

Mark Schemes begin on Page 8

Question 1 | Source: 9618_w25_qp_12 (Q5.e)

5 A local area network (LAN) has five computers, one switch and one server.

(e) Complete the table by identifying **one** threat to computer and data security posed by networks and the internet.

Describe the threat and give a method of prevention.

[illegible]

[3]

Question 2 | Source: 9618_s25_qp_11 (Q7.b)

7 A banker stores personal data on their work computer.

(b) The banker receives confidential data across the internet. The data includes a digital signature.

Explain how a digital signature can make sure the data has **not** been changed during transmission.

[illegible]

[5]

Question 3 | Source: 9618_s25_qp_12 (Q5.c.i)

- 5 An online game has a database that stores data about users and the characters each user creates in the game. Each user can create multiple characters and purchase multiple items for each character.

The normalised database has the following design:

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USER (Username, Password, DateOfBirth)

CHARACTER (CharacterName, CharacterID, Username, Level, Money)

ITEM (ItemName, MinimumLevel, Cost)

CHARACTER_ITEM (CharacterID, ItemName)
```

(c) A Database Management System (DBMS) provides data security.

- (i) Identify **two** methods the DBMS can use to protect the data in the table `USER` from unauthorised access.

Explain how each method protects the data.

Method 1

Explanation

.....

.....

Method 2

Explanation

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.....

[4]

Question 4 | Source: 9618_s24_qp_11 (Q5.c.i)

- 5 A bank allows customers to access their accounts using an application that they can download onto a device such as a smartphone.

(c) The bank also needs to keep its customers' data private and secure.

- (i) The bank's network has a firewall.

Explain how a firewall can help protect the customers' data.

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..... [3]

Question 5 | Source: 9618_s24_qp_12 (Q3.a.i)

- 3 An assessment board scans exam papers and stores the digitised papers on a server. Exam markers download the digitised papers to mark. The exam markers then upload the mark for each paper.

(a) The assessment board needs to make sure the data stored on the server is secure.

- (i) Authentication methods can help to protect the server against hackers.

Identify **one other** security measure that helps to protect the server from hackers.

Describe how the security measure works.

Security measure

Description

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..... [3]

Question 6 | Source: 9618_w23_qp_13 (Q3.a.iii)

- 3 A company sells online Computer Science courses to students in different countries.

The courses are stored on a public cloud.

(iii) State how the following security measures can be used to protect computer systems.

Firewall

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Encryption

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Passwords

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[3]

Question 7 | Source: 9618_s23_qp_13 (Q6.a)

6 Data needs to be kept secure when stored on a computer and during transmission over a network.

(a) Explain how a digital signature is used to authenticate a digital document during transmission over a network.

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..... [5]

Question 8 | Source: 9618_s23_qp_13 (Q6.b)

6 Data needs to be kept secure when stored on a computer and during transmission over a network.

- (b) Complete the table by identifying **and** describing **two** types of software that can be installed on a computer to prevent threats over a network.

Type of software	Description
.....	
.....	

[2]

Question 9 | Source: 9618_w22_qp_11 (Q2)

- 2 Draw **one** line from each security feature to its most appropriate description.

Security feature	Description
firewall	converts data to an alternative form
pharming	redirects a user to a fake website
anti-virus software	verifies the authenticity of data
encryption	scans files on the hard drive for malicious software
	accepts or rejects incoming and outgoing packets based on criteria

[4]

Question 10 | Source: 9618_w22_qp_12 (Q6.a.i)

- 6 (a) A student uses a networked laptop computer to send an email to a colleague.

- (i) Explain how a digital signature ensures the email is authentic.

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..... [2]

Question 11 | Source: 9618_w22_qp_12 (Q6.a.ii)

- 6 (a) A student uses a networked laptop computer to send an email to a colleague.

- (ii) Describe how a firewall protects the data on the computer.

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..... [3]

Question 12 | Source: 9618_w21_qp_12 (Q1)

- 1 When designing computer systems, it is important to consider the security, integrity and privacy of the data.

Draw **one** line from each measure to indicate whether it keeps data secure or protects the integrity of data.

Measure

Firewall

Double entry

Presence check

Access rights

Password

Data Security

Data Integrity

[2]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_12 (Q5.e)

5(e)	1 mark for correct threat 1 mark corresponding description 1 mark for correct method of prevention <table border="1"> <thead> <tr> <th>Threat</th><th>Description</th><th>Prevention</th></tr> </thead> <tbody> <tr> <td>Malware / Virus</td><td>Malicious code that can alter/delete files</td><td>Anti-virus // Anti-malware // Firewall</td></tr> <tr> <td>Spyware</td><td>Records keystrokes which are sent to third party</td><td>Anti-spyware // Firewall // Anti-malware</td></tr> <tr> <td>Hacking</td><td>Gaining <u>unauthorised</u> access to a computer network / device</td><td>Authentication // Firewall</td></tr> <tr> <td>Phishing</td><td><u>Emails</u> supposedly from reputable companies are sent to trick people into revealing personal information</td><td>Use a spam filter // Do not open emails from unknown sources</td></tr> <tr> <td>Pharming</td><td>Users are directed to a bogus <u>website</u> that looks legitimate to obtain personal information</td><td>Use a VPN // Anti-malware// Do not open links or download attachments</td></tr> </tbody> </table>	Threat	Description	Prevention	Malware / Virus	Malicious code that can alter/delete files	Anti-virus // Anti-malware // Firewall	Spyware	Records keystrokes which are sent to third party	Anti-spyware // Firewall // Anti-malware	Hacking	Gaining <u>unauthorised</u> access to a computer network / device	Authentication // Firewall	Phishing	<u>Emails</u> supposedly from reputable companies are sent to trick people into revealing personal information	Use a spam filter // Do not open emails from unknown sources	Pharming	Users are directed to a bogus <u>website</u> that looks legitimate to obtain personal information	Use a VPN // Anti-malware// Do not open links or download attachments	3
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Mark Scheme for Question 2 | Source: 9618_s25_ms_11 (Q7.b)

7(b)	1 mark each to max 5 • The sender hashes the document / message • to produce a digest • The sender encrypts the digest to create the digital signature • The message and the signature are sent to the banker / receiver • The receiver decrypts the signature to reproduce the digest • The receiver uses the same hashing algorithm on the document received to produce a second digest • The receiver compares this digest with the one from the digital signature • If both of the receiver's digests are the same the document has not changed	5
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Mark Scheme for Question 3 | Source: 9618_s25_ms_12 (Q5.c.i)

5(c)(i)	1 mark for method 1 mark for matching explanation e.g. • Access rights • appropriate permissions for the table USER are needed to read or edit the data • A password for the database or the USER table • prevents users without the password from accessing the data • Encrypting the database • stops users without the decryption key from decoding / understanding the data • Views • users can be given a view to the database that does not include the data in the table USER	4
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Mark Scheme for Question 4 | Source: 9618_s24_ms_11 (Q5.c.i)

5(c)(i)	1 mark each to max 3 : • Compares all incoming and outgoing transmissions • ... against set criteria/whitelist/blacklist • Blocks all transmissions that do not meet rules • Blocks data entering from specific ports • Blocks unauthorised/unknown internal software transmitting data	3
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Mark Scheme for Question 5 | Source: 9618_s24_ms_12 (Q3.a.i)

3(a)(i)	1 mark for security measure 1 mark each to max 2 for how the chosen measure works: • Firewall • Checks incoming connections • ... against criteria • Blocks data from entering specific ports • Blocks data that does not meet whitelist that meets blacklist • Proxy server • Prevents devices accessing the web server directly • Intercepts any requests • Forwards the request using its own IP address • Screens returning data before sending it to the user	3
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Mark Scheme for Question 6 | Source: 9618_w23_ms_13 (Q3.a.iii)

3(a)(iii)	1 mark each for firewall, encryption and passwords. Firewall: - Monitors incoming and outgoing traffic and rejects any traffic that does not meet the set rules Encryption: - Ensures that if data is intercepted / obtained it cannot be understood without the decryption key Passwords: - Ensures only users with the correct password can access the resources // prevents unauthorised access	3
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Mark Scheme for Question 7 | Source: 9618_s23_ms_13 (Q6.a)

6(a)	1 mark each to max 5 - The sender hashes the document ... to produce a <u>digest</u> - The sender <u>encrypts</u> the digest to create the digital signature - The message and the signature are sent to the receiver - The receiver <u>decrypts</u> the signature to reproduce the digest - The receiver uses the <u>same</u> hashing algorithm on the document received to produce a second digest - The receiver compares this digest with the one from the digital signature - If both of the receiver's digests are the same the document is authentic	5
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Mark Scheme for Question 8 | Source: 9618_s23_ms_13 (Q6.b)

6(b)

1 mark each for identification and appropriate description of 2 pieces of software, **max 2**

2

Type of software	Description
Antivirus	- scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing
Antispyware	- scans the computer for spyware and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known spyware and prevents the download continuing.
Firewall	- monitors incoming and outgoing traffic and compares it to criteria that are set by the user such as through a whitelist/blacklist/identifying allowed / blocked IP addresses - compares incoming and outgoing traffic to criteria blocks those that do not match criteria
Antimalware	- scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing

Mark Scheme for Question 9 | Source: 9618_w22_ms_11 (Q2)

2	1 mark for each correct line. Security feature firewall pharming anti-virus software encryption converts data to an alternative form redirects a user to a false website verifies the authenticity of data scans files on the hard drive for malicious software accepts or rejects incoming and outgoing packets based on criteria Description	4
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Mark Scheme for Question 10 | Source: 9618_w22_ms_12 (Q6.a.i)

6(a)(i)	1 mark for each method of ensuring authenticity (max 2): • (email) message put through hashing algorithm to produce a digest • Digest encrypted with <u>sender's private</u> key (to create the digital signature) • the (digital) signature can only be decrypted with matching <u>sender's public</u> key	2
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Mark Scheme for Question 11 | Source: 9618_w22_ms_12 (Q6.a.ii)

6(a)(ii)	1 mark for each bullet point: • monitors <u>incoming</u> and <u>outgoing</u> packets / traffic • checks against an allow list / deny list of IP addresses // checks against a set of rules for acceptable data / ports etc. • blocks transmissions that do not meet criteria / rules // allows through is satisfies the criteria /rules	3
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Mark Scheme for Question 12 | Source: 9618_w21_ms_12 (Q1)

1	1 mark for 3 correct lines only from Data Security 1 mark for 2 correct lines only from Data Integrity <pre>graph LR; Firewall[Firewall] --- DS[Data Security]; DoubleEntry[Double entry] --- DS; PresenceCheck[Presence check] --- DI[Data Integrity]; AccessRights[Access rights] --- DS; Password[Password] --- DI;</pre>	2
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